DocuTrace Verification Engine

Document Comparison & Modification Analysis Report

Generated: 2026-10-15 07:49:12 UTC | Report ID: **4b7b49dc89f8** | Engine: pdf-diff + PyMuPDF + OpenCV + DAA

1. Executive Summary & Verdict

Overall Assessment Verdict	**CRITICAL DATA MODIFIED**	**Total Changes Detected**	**20**
LCS Text Similarity	88.2%	Text Add/Del/Replace	11
Edit Similarity (Levenshtein)	89.3%	Numerical / Date Changes	5
Section Inversion Score	0.0 (0 inversions)	Font / Visual Changes	8 font / 1 diagram

2. Document Information

Attribute	**Original Document (Source)**	**Modified Document (Target)**
Filename	07_research_paper_original.pdf	08_research_paper_modified.pdf
Page Count	1	1
File Size	2.6 KB	2.8 KB
SHA-256 Hash	3a2425b0f97404385503429465082433...	f4b334be31b417f8403aaec864413ea6...

3. Modified Research Paper Sections

Section Name	**Status**	**Changes**	**Words Added**	**Words Removed**	**Key Evidence**
Abstract	MODIFIED	1	2	1	Changed: '98.4%.' -> 'of 72.1%.'
Introduction	MODIFIED	1	0	0	Figure or diagram graphic modification on page 1 (...
Methodology	MODIFIED	3	12	1	Added: 'imagery.'; Added: 'An additional experimen...
Results	MODIFIED	1	2	1	Changed: '98.4%' -> 'of 72.1%'
Conclusion	MODIFIED	1	3	2	Changed: 'Original DAA' -> '1: Modified Pipeline'
General Content	MODIFIED	13	9	6	Changed: 'Ada Lovelace' -> 'Dr. John von Neumann';...

4. Granular Detected Modifications

Pg	**Section**	**Category**	**Original Text**	**Modified Text**	**Change Details**
1	General Content	NAME_ENTITY	Ada Lovelace	Dr. John von Neumann	Modified in General Content: 'Ada L…
1	General Content	REGISTRATION	ARXIV-2024-8841	ARXIV-2024-9999	Modified in General Content: 'ARXIV…
1	General Content	DATE	October	Published: November	Modified in General Content: 'Octob…
1	General Content	NUMERICAL	15,	20,	Modified in General Content: '15,' …
1	General Content	NUMERICAL	2024	2025	Modified in General Content: '2024'…
1	Abstract	NUMERICAL	98.4%.	of 72.1%.	Modified in Abstract: '98.4%.' -> '…
1	Methodology	TEXT		imagery.	Text insert in Methodology: 'imager…
1	Methodology	TEXT		An additional experimenta…	Text insert in Methodology: 'An add…

1	Methodology	TEXT	imagery.		Text delete in Methodology: 'imager…
1	Results	NUMERICAL	98.4%	of 72.1%	Modified in Results: '98.4%' -> 'of…
1	Conclusion	TEXT	Original DAA	1: Modified Pipeline	Modified in Conclusion: 'Original D…
1	General Content	FONT_ONLY	We present a robust verif…	We present a robust verif…	Typographic change in General Conte…
1	General Content	FONT_ONLY	Document tampering repres…	Document tampering repres…	Typographic change in General Conte…
1	General Content	FONT_ONLY	Traditional text-only ext…	Traditional text-only ext…	Typographic change in General Conte…
1	General Content	FONT_ONLY	Our architecture combines…	Our architecture combines…	Typographic change in General Conte…
1	General Content	FONT_ONLY	Computer vision differenc…	Computer vision differenc…	Typographic change in General Conte…
1	General Content	FONT_ONLY	4. Results	4. Results	Typographic change in General Conte…
1	General Content	FONT_ONLY	5. Conclusion	5. Conclusion	Typographic change in General Conte…
1	General Content	FONT_ONLY	The integrated pipeline e…	The integrated pipeline e…	Typographic change in General Conte…
1	Introduction	VISUAL_DIAGR			Figure or diagram graphic modificat…

5. Font & Formatting Differences

Page	Section	Text Content	Formatting Differences Detected
1	General Content	We present a robust verification en	font: Helvetica -> Times-Roman, size: 10.0 -> 11.0, color: #
1	General Content	Document tampering represents a sub	font: Helvetica -> Times-Roman, size: 10.0 -> 11.0, color: #
1	General Content	Traditional text-only extraction fr	font: Helvetica -> Times-Roman, size: 10.0 -> 11.0, color: #
1	General Content	Our architecture combines the open-	font: Helvetica -> Times-Roman, size: 10.0 -> 11.0, color: #
1	General Content	Computer vision differencing with O	font: Helvetica -> Times-Roman, size: 10.0 -> 11.0, color: #
1	General Content	4. Results	position: None -> None
1	General Content	5. Conclusion	position: None -> None
1	General Content	The integrated pipeline effectively	font: Helvetica -> Times-Roman, size: 10.0 -> 11.0, color: #

6. Visual Figures, Diagrams & Image Modifications (OpenCV)

Page	Kind	Severity	Area (px)	Description
1	FIGURE	SIGNIFICANT	51816	Figure or diagram graphic modification on page 1

7. DAA Algorithmic Verification Details

Algorithm	Role / Target	Time (ms)
pdf-diff Bounding Box Alignment	Text layer alignment & bounding box delta extraction	9.1 ms
PyMuPDF Span Typography Engine	Font family, size, style flag, and hex color comparison	6.93 ms
OpenCV Visual Differencing	Page rasterization, contour detection & figure diffing	30.73 ms
LCS Dynamic Programming (DAA)	Token-level longest common subsequence alignment	2.26 ms
Levenshtein Distance with Operations	Character-level substitution, insertion, deletion scripts	292.13 ms
Inversion Counting Algorithm (DAA)	Quantifying section order permutation & document reordering	<1 ms

Limitations & Compliance: This document comparison and modification analysis report was generated by deterministic computer algorithms combining the open-source pdf-diff framework (CC0 1.0), PyMuPDF layout analysis, OpenCV visual computer vision, and DocuTrace DAA algorithms. It measures textual, typographic, and graphic variance. It does not constitute a legal determination of document authenticity, validity, or authorial intent. A

qualified reviewer must inspect all highlighted evidence.